



Time: 5 minutes



Outcome: Calculate and interpret output losses associated with disinflation.

Difficulty: ●●○ Intermediate

Sacrifice Ratio



THE CORE IDEA

The sacrifice ratio is cumulative output lost, as a percentage of one year's potential output, per percentage-point reduction in inflation. Add annual percentage output gaps before dividing by the inflation decline. Credibility and faster expectations adjustment can reduce the cost of disinflation.



HOW TO RECOGNIZE IT

- Use percentage output gaps, not unscaled dollar losses.
- Add losses across all years of the disinflation.
- Divide by the percentage-point fall in inflation.



WATCH OUT

A common mistake is to divide the dollar output loss directly by the inflation change. First express the loss as a percentage of potential output, then divide by the percentage-point reduction in inflation.



WORKED EXAMPLE: CALCULATING THE SACRIFICE RATIO

Disinflation output losses (billions of dollars)

	Potential	Actual	Output loss	Gap / potential
Year 1	400	388	12	3%
Year 2	400	388	12	3%
Total loss	—	—	24	6% of one year

Each year loses \$12 billion relative to \$400 billion potential output, a 3% gap. Add the two annual gaps: cumulative loss is 6% of one year's potential output. Inflation falls by 3 percentage points, so the sacrifice ratio is $6 / 3 = 2$. Keep percentage output losses distinct from percentage-point inflation changes.



CHECK YOURSELF

Should annual output gaps be added before calculating a multi-year sacrifice ratio?



READY?

Return to the game and master this concept.